Example of matrix 3×3

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1(45-48) - 4(18-24) + 7(12-15) = (-3) - 4(-6) + 7(-3)$$

$$= -3 + 24 - 21 = 0$$

$$\begin{vmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{vmatrix}$$

$$45 + 84 + 96 - 105 - 48 - 72 = 225 - 225 = 0$$

$$\begin{vmatrix} (45-48) & (18-24) & (12-15) \\ (36)-(42) & (9-21) & (6-12) \\ (32-35) & (8-14) & (5-8) \end{vmatrix} = \begin{vmatrix} -3 & -6 & -3 \\ -6 & -12 & -6 \\ -3 & -6 & -3 \end{vmatrix}$$

singular

---

non singular case

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 4 \end{vmatrix} = 1(8-1) - 3(8-3) + 4(2-6) = (7) - 3(5) \blacksquare 16$$

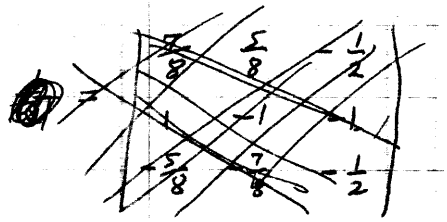
$$= 1(7) + 3(-5) + 4(-4) = \blacksquare\ 7 - 31 = -24$$

$$8 + 8 + 9 - 24 - 1 - 24 = 25 - 49 = -24$$

$$\begin{vmatrix} 1 & 3 & 4 \\ 2 & 2 & 1 \\ 3 & 1 & 4 \end{vmatrix}$$

$\frac{7}{8} + 3 - \frac{21}{8}$

$$\frac{1}{24}\begin{vmatrix} (8-1) & -(8-3) & (2-6) \\ (12-4) & (4-12) & -(1-9) \\ (3-8) & -(1-8) & (2-6) \end{vmatrix} = \frac{1}{24}\begin{vmatrix} 7 & -5 & -4 \\ -8 & -8 & +8 \\ -5 & +7 & -4 \end{vmatrix} =$$



$$\begin{vmatrix} \frac{7}{8} & \frac{5}{8} & -\frac{1}{2} \\ 1 & -1 & -1 \\ -\frac{5}{8} & \frac{7}{8} & -\frac{1}{2} \end{vmatrix} \cdot \begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 4 \end{vmatrix} = \begin{vmatrix} \frac{7}{8}+\frac{15}{8}-2 & & \\ 1-3-4 & & \\ -\frac{5}{8}-\frac{21}{8}-2 & & \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 - 6 = +2$$

$$\begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix}$$

$$\begin{vmatrix} 4 & 2 \\ 3 & 1 \end{vmatrix} = \begin{vmatrix} +2 & -1 \\ -\frac{3}{2} & +\frac{1}{2} \end{vmatrix}$$

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \cdot \begin{vmatrix} +2 & -1 \\ -\frac{3}{2} & +\frac{1}{2} \end{vmatrix} = \begin{vmatrix} 2-3 & -1+1 \\ 6-6 & -3+2 \end{vmatrix} = \begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix}$$

multiply back:

$$\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 4 \end{vmatrix} \cdot \begin{vmatrix} \frac{7}{8} & -\frac{5}{8} & \frac{1}{2} \\ -1 & -1 & +1 \\ -\frac{5}{8} & \frac{7}{8} & -\frac{1}{2} \end{vmatrix} = \begin{vmatrix} \frac{7}{8}-2-\frac{15}{8} & -\frac{5}{8}-2+\frac{21}{8} & \\ \frac{21}{8}-2-\frac{5}{8} & -\frac{15}{8}-2+\frac{7}{8} & \\ \frac{7}{2}+1-\frac{5}{2} & & \end{vmatrix}$$

$$\frac{1}{-24}\begin{vmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 4 \end{vmatrix} \cdot \begin{vmatrix} 7 & -5 & -4 \\ -8 & -8 & +8 \\ -5 & +7 & -4 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$= \frac{1}{-24}\begin{vmatrix} (7-16-15) & (-5-16+21) & (-4+6-12) \\ (21-16-5) & (-15-16+7) & (-12+16-4) \\ (28-8-20) & (-20-8+28) & (-16+8-16) \end{vmatrix} = \frac{1}{-24}\begin{vmatrix} -24 & 0 & 0 \\ 0 & -24 & 0 \\ 0 & 0 & -24 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

2×2 case

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{vmatrix}$$

$$\begin{vmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \cdot \begin{vmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{vmatrix} = \begin{vmatrix} a_{11}a_{22}-a_{12}a_{21} & -a_{12}a_{11}+a_{12}a_{11} \\ a_{21}a_{22}-a_{21}a_{22} & -a_{12}a_{21}+a_{11}a_{22} \end{vmatrix}$$

$$= \begin{vmatrix} a_{11}a_{22}-a_{12}a_{21} & 0 \\ 0 & a_{11}a_{22}-a_{12}a_{21} \end{vmatrix} = |A| \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

determinant:

$$\begin{vmatrix} 1 & 2 & 3 \\ & & \end{vmatrix} \cdot \begin{vmatrix} 7 \\ -8 \\ -5 \end{vmatrix} \quad \frac{1}{-24}(7-16-15)$$